

STM32 Resource Budget B

Resource	Required	Signals
ADC Inputs	8	3 voltage + 3 current + CP voltage + temperature
EXTI Inputs	1	RCM fault
Timer PWM	1	CP PWM
UART/USART	2	TX + RX
SPI Alternate Functions	3	SCK + MOSI + MISO
SPI Chip Select	1	RFID CS/NSS
Normal GPIO Inputs	2	Contactactor 1 + Contactactor 2 feedback
Normal GPIO Outputs	9	2 contactor controls + buzzer + 6 LCD
SWD	2	SWDIO + SWCLK

Recommended phys

ADC → CP Timer PWM → SPI RFID → UART → RCM EXTI →

Before Physical Pin Assignment

Assignment Note

Assign first because only ADC-capable GPIOs can be used.

RCM gets interrupt capability for fast fault response.

Requires a pin supporting a suitable TIMx_CHx alternate function.

Both pins must map to the same selected UART/USART peripheral.

All three must map to the same selected SPI peripheral.

Can normally be a GPIO output; hardware NSS is optional depending on design.

Periodic GPIO reading is sufficient unless later requirements demand interrupts.

Flexible pins; assign after constrained peripherals. RFID CS is counted separately above.

Reserve for development/programming/debugging.

ical pin assignment order

Contact I/O → Buzzer/LCD GPIO → verify SWD remains reserved